

Operating instructions

Ambulatory 24-hour blood pressure monitoring system

MOBIL  **GRAPH**®
New **G**eneration 24h ABP-CONTROL

CE 0044



Blood pressure monitor

Evaluation units:

HMS for WINDOWS® Version 7.x

Dealer:



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1. INTRODUCTION

1.1 Foreword

This Mobil-O-Graph NG® 24-hour **A**mbulatory **B**lood **P**ressure **M**onitor system (ABPM) provides you with a system using the latest specialist technology, which incorporates all our experience of the past few years with the successful Mobil-O-Graph® ABP monitor.

We have also incorporated our customers' wishes into the Mobil-O-Graph NG®.

The blood pressure monitor, also referred to below as the ABP monitor or recorder, can be prepared for a new patient at the surgery in just a few minutes. This gives you the advantage of using the ABP monitor optimally, and the ability to complete a whole monitoring session each day. The Mobil-O-Graph NG® will thus quickly be integrated into the daily routine of the practice.

Clinical experience has shown that in addition to the ABPM, analysis of blood pressure self-monitoring is also sensible. The client/server data management extension enables a virtual patient clinic to be established.

1.2 The operating instructions

These operating instructions will show you quickly and simply how to use the ABP monitoring system and its accessories. With a little practice you will see how easy it is to use the ambulatory blood pressure monitor.

The Mobil-O-Graph NG® is supplied by I.E.M. in two versions with different evaluation programs:

- Hypertension Management Software for Windows (from Version 7.0).

Where reference is made to specific characteristics, only those sections relating to your version are relevant!

Important: The operating instructions describe the ABP monitor and its accessories in the sequence in which you will be setting up the equipment and then using it. After the general description you will find details of installation, setting up, preparation for monitoring, fitting to the patient and evaluation. The individual functions are explained later, at the point at which they are to be used. You will therefore become familiar with use of the blood pressure monitor step by step.

1.3 Clinical tests

The Mobil-O-Graph NG® blood pressure monitor fulfils the requirements of the ESH (European Society of Hypertension).

1.4 CE mark



The Mobil-O-Graph NG® fulfils the requirements of the Medical Products Directive 93/42/EEC and bears the CE mark.

The Mobil-O-Graph NG® fulfils the requirements of the R&TTE Directive 1999/5/EU and bears the CE mark.

The declaration of conformity for this product can be downloaded from the I.E.M. website www.iem.de.

This device fulfils the requirements of the standards applicable in accordance with the Directives. In particular, it fulfils the applicable requirements of the following standards:

1. EN 1060-1 Noninvasive blood pressure measuring equipment – Part 1: General requirements
2. EN 1060-3 Noninvasive blood pressure measuring equipment – Part 3: Supplementary requirements for electromechanical blood pressure measuring systems

1.5 Scope of delivery

Basic set:

1. ABP monitor Mobil-O-Graph NG®
2. S, M und L Cuffs
3. Cuff tube, incl. connector
4. Pouch for Mobil-O-Graph NG®
5. Charger, incl. 4 NiMH batteries
6. Operating instructions
7. Measuring tape

Analysis options:

Analysis unit HMS for Windows®

1. HMS software on CD
2. USB/serial PC cable
3. IR-MED serial or USB (optional)

2. Operating instructions

2.1 Correct use

The correct use is the 24-hour blood pressure monitoring of adults. The Mobil-O-Graph NG® should be used only under medical supervision.

2.2 Incorrect use

Because of the risk of strangulation from the tube and cuff, the ABPM System Mobil-O-Graph NG® may not be used for patients who are legally incompetent, and may not get into the hands of unsupervised children.

The Mobil-O-Graph NG® may not be used for blood pressure monitoring in intensive care or during operations.

2.3 Indication

The Mobil-O-Graph NG® ABPM system is intended for clarifying blood pressure and for use as a diagnostic aid. The doctor can, at his or her own discretion, prescribe the use of this medical system for patients if they are suffering from:

- hypertension,
- hypotension,
- borderline hypertrophy
- impaired renal function, or
- require antihypertensive therapy.

2.4 Side effects of long-term blood pressure monitoring

Long-term blood pressure monitoring is now a familiar, acceptable and valuable method of measurement, used daily in diagnostic investigation and treatment follow-up.

When considering the use of long-term blood pressure monitoring, it is important to check whether the patient is suffering from coagulation disorders or is on anticoagulant treatment. As with occasional blood pressure measurement, petechial bleeding can occur.

In patients on anticoagulant therapy or patients with coagulation disorders, petechial bleeding can occur on the measurement arm in spite of correct cuff fit. The special risk to patients due to anticoagulant treatment or coagulation disorders is present in all cases irrespective of the type of equipment used in measurement.

2.5 Important instructions for the doctor

- A measurement can be broken off at any time by pressing one of the monitor keys. This deflates the cuff. (except for the Day/Night key)
- Inform your patients, that if a measurement should ever become painful, they should switch off the unit and take off the cuff.
- Explain to the patient how to place the cuff correctly.
- The equipment must not be used in the vicinity of an MRI scanner!
- The equipment must never be connected to a PC or other unit while still fitted to the patient.
- The use of components not supplied with the equipment can lead to incorrect readings.
- The air tube between the Mobil-O-Graph NG® recorder and the cuff must never be knotted, compressed or stretched.
- Observe the hygiene rules shown in the maintenance schedule.

Warnings:

- The shoulder strap or cuff tube can become entangled around the patient's neck and lead to strangulation. Countermeasures: The doctor should draw the patient's attention to the fact that the cuff must be worn on the upper arm only and care must always be taken to ensure that neither the shoulder strap nor the air tube can ever become wrapped around the neck. The air tube should therefore always be worn under the outer clothing (even at night).
- If you suspect that water may have entered the unit during cleaning or use of the equipment, the device may no longer be used on the patient. Inform your service engineer without delay or send it for testing.
- The air tube may kink when inflated. The doctor should explain to the patient that, particularly when sleeping, the equipment should be positioned in such a way that the tube cannot be compressed. If the patient is not fully competent, the equipment should be worn only under supervision.
- In some patients, petechia, haemorrhages or subcutaneous haematomas are possible. All patients must be told when putting on the cuff that if they experience pain they should switch off the equipment and inform the doctor.
- The doctor must be convinced that, according to the health of the patient, the use of the device will not damage blood circulation in the arm.

- The Mobil-O-Graph NG® is not suitable for use at the same time as high-frequency surgical equipment.
- The Mobil-O-Graph NG® fulfils the requirements of EMC standards; nevertheless, the Mobil-O-Graph NG® should not be exposed to strong electromagnetic fields, as outside the limits they may cause malfunction.
- The cuffs and the tube are made of electrically nonconductive material. They thus protect the device from the effects of a defibrillator discharge.
- During a defibrillator discharge the device must not be allowed to touch the patient. The device may be damaged by such a discharge and give inaccurate readings.

2.6 Important instructions for the patient

- The beep signals a measurement. Keep still during the measurement until the process is complete. Let your arm hang loosely or, when sitting, let your lower arm rest on the table or on a support. Avoid moving your hand during the measurement.
- The Start key  triggers an additional measurement. Press this key only if instructed to do so by your doctor (e.g. after taking medication, if you have chest pains or shortness of breath, etc.).
- The Event key  saves the time of an event and triggers a measurement. Press this key only if something important or unusual is happening.
- The Day/Night key  should be pressed immediately before going to sleep and upon waking.
- The ON/OFF key  permits the device to be switched on and off if a measurement is not desired at the set time. To switch off, press the key for longer than 2 seconds.
- A measurement can be broken off at any time by pressing any of the keys. This deflates the cuff.

Warnings:

- The cuff may only be worn on the upper arm and in every case you should ensure that neither the shoulder strap nor the air tube can ever become wrapped around your neck. The air tube should therefore always be worn under the outer clothing (even at night).
- Take care to ensure that the tube of the cuff cannot be kinked or compressed, particularly while sleeping.
- If the measurement causes you pain, switch off the device, take off the cuff and inform your doctor.
- The blood pressure monitor must not come into contact with liquids. Do not wear the equipment in the shower! If, for example, water has entered the device, it may no longer be used. Switch off the device and take out the batteries.
- Serious malfunctions are indicated by a continuous beep. If a continuous beep sounds, switch off the device, take off the cuff and inform your doctor.
- The cuff connection (air connector) must always engage with an audible click. Otherwise there will be a leaky connection between the Mobil-O-Graph NG® and the tube that will produce incorrect measurements.

You can find a form to copy and give to the patient in the Appendix!

3. Product description

3.1 Introduction

The Mobil-O-Graph NG® system consists of two principal components:

- Mobil-O-Graph NG® Basic set with various cuffs and further accessories
- Interpretation software for the doctor to evaluate the readings.

Readings from the ABP monitor can be interpreted using the Analysis unit (either: HM Client software or HM Windows software). Using the software, saved readings can be downloaded to a PC, presented on the screen in various forms, as diagrams, lists and statistics, and printed out.

The Mobil-O-Graph NG® can then be prepared immediately for the next patient. With little practice, this procedure can be completed in just a few minutes. This allows the doctor to use the equipment on each working day, around the clock.

The Mobil-O-Graph NG® monitoring system is designed to allow the recording and display of a blood pressure profile throughout the day and night. Additional parameters are detected, such as nocturnal values and blood pressure variations. This allows the doctor to prescribe the optimum medication for the individual and to monitor the therapeutic outcome.

3.2 Unpacking

All parts supplied are packed carefully for despatch and checked for completeness and correct function. Should the goods supplied be incomplete or damaged, please notify the supplier without delay.

3.3 The monitor

Components:

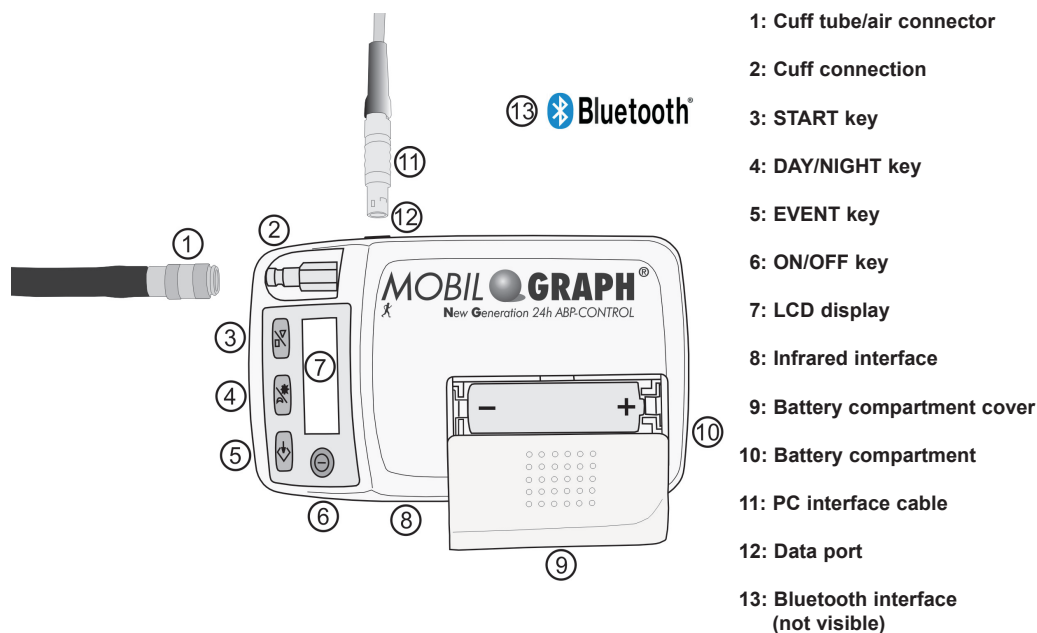


Figure 3.1 Monitor, top view

3.3.1 Keys

All keys are located on the front of the recorder housing (see Figure 3.1 Monitor, top view).

ON/OFF This key switches the ABP recorder on and off. To prevent switching it on or off by mistake, the key reacts only after 2 seconds. As with all the other keys, blood pressure measurement can also be interrupted prematurely, causing the pressure in the cuff to be released rapidly (see also section Warnings).
Caution: Switch the unit on again to continue.



DAY/NIGHT The DAY/NIGHT key allows waking and sleeping phases to be differentiated when recording, which is important from the point of view of statistical evaluation and graphic presentation. For precise details of the effects on the printout, please see the relevant sections on analysis.



Briefly: the patient is instructed to operate the "DAY/NIGHT" key on going to bed and again when getting up in the morning. This adjusts the monitoring interval to suit the individual patient and helps you in assessing the blood pressure profile. As well as adjusting the interval, the appropriate notes are shown on the printout. If this key is not operated, the change in interval will take place according to the protocol selected.

EVENT

This key allows the patient to record the time at which medication is taken or any events that may cause blood pressure to rise or fall. Pressing this key directly triggers a measurement. The reason for pressing the Event key should be noted by the patient in the events diary.

START

This key starts the 24-hour measurement period and also allows a measurement outside the pre-set measurement cycle.

Caution:

The result of the first measurement should be checked by the doctor for plausibility, to ensure that the subsequent automatic measurements are correct, and that the cuff fits properly. If the measurement is incorrect, the instructions in the sections Preparing for Monitoring and Troubleshooting should be followed. When the Start key is pressed, the number of measurements recorded so far is displayed and manual measurement begins. This differs from automatic measurement according to the monitoring protocol, in that the cuff is inflated in stages. This determines the cuff pressure required to allow the systolic blood pressure to be measured. This maximum inflation pressure is stored and is then approached immediately by direct inflation for the subsequent automatic measurements. The patient can use the Start key to take manual measurements in addition to the pre-set measurement cycle.

3.3.3 Display

The LCD display is located on the front of the recorder housing (see Figure 5.1 Monitor, top view). It displays useful information for the doctor and patient on measurement data, monitor settings and measurement errors.

3.3.4 Acoustic signals

The acoustic signals used consist of single or multiple beeps.
The following signals are emitted:

- 1 beep: Switching on or off, start and completion of a measurement (except for night intervals), removal of interface cable, end of IR communication, set-up and end of Bluetooth communication, and measuring error
- 3 beeps: System error
- Permanent beep: Serious system error (e.g. cuff pressure is greater than 15 mmHg for more than 10 seconds outside of a measurement)
- Combined beeps: For manual deletion of readings, first one beep is emitted and 2 seconds later 5 beeps.

3.3.5 Cuff connection

The cuff connection is fitted on the front of the recorder housing (see Figure 5.1 Monitor, top view). This metal plug is used to connect the recorder with the cuff via the cuff tube and the metal cuff socket.

Important: Both the doctor and the patient should be aware that the metal plug (air connector) must always engage with an audible click. Otherwise there will be a leaky connection between the Mobil-O-Graph NG® and the tube that will produce incorrect measurements.

3.3.6 Data port

The data port is located on the left-hand side of the housing (see Figure 3.1 Monitor, top view). The connector cable supplied goes into this socket. This is a plug-in connection, i.e. the red dot on the plug must be aligned with the red dot on the socket. To disconnect, pull on the outer metal ring of the plug.
(see section The analysis unit)

3.3.7 The infrared interface

The infrared interface provides an equal, wireless alternative to the data port. To use this interface you will need the PC infrared interface IR-Med. The infrared interface can be obtained from your specialist dealer.
(see section The analysis unit)

3.4 Preparing for monitoring

- Connect the cuff tube with one of the cuffs supplied by sticking it onto the plastic connector.
- Connect the cuff tube with the socket on the front of the recorder housing.
- First check that the batteries have been inserted correctly. You should always use fully charged batteries for a new monitoring session. Alternatively, you can use alkaline batteries. Take care to insert the batteries the right way round.

IMPORTANT:

Use only the I.E.M. NiMH rechargeable batteries supplied or alkaline batteries. Although zinc-carbon batteries and NiCd rechargeables show adequate voltage in the battery test, the output is often insufficient to carry out monitoring over a 24-hour period.

The rechargeable batteries should be discharged and recharged several times before they are first used. Please follow the instructions for use enclosed with the charging unit.

To perform monitoring over a 48-hour period, you need 2 additional batteries to exchange after 24 hours.

3.4.1 Switching on

Always check the condition of your recorder first, before you give it to a patient. You can do this by observing the initial displays shown on the recorder shortly after switching it on. The following sequence of displays should be visible:

Test	Display	Commentary
Battery status (Volts):	2.85	(for NiMH rechargeable batteries min. 2.75; for alkaline batteries min. 3.10)
Display Segment Test	999:999 to 000:000	Along with the numbers, all the other LCD symbols are shown. Check whether all segments are being shown correctly and completely (the complete code is constantly being checked for correctness in the background)
Current 24-hour time:	21:45	(Example)

If a fault occurs during the internal test, the recorder shows **E004** on the display and emits an acoustic signal. For safety reasons, use of the recorder is blocked. Return the recorder to your specialist dealer or direct to IEM GmbH for repair without delay.

3.4.2 Clearing the memory

The memory must be cleared before each monitoring session, i.e. no blood pressure data from the previous patient must be left in the memory. However, should values still be present, you can delete them with the delete function of the software.

You can delete the memory manually by pressing and holding down the start button for at least 5 seconds. While holding down, all segments of the LCD are highlighted followed by a single beep. Subsequently the number of the recordings is shown, followed by "clr". If the event button is pressed within 5 seconds and held down for at least 2 seconds, all recordings will be deleted.

3.4.3 Setting the time/date

The Mobil-O-Graph NG® has an internal buffer battery that allows the clock to continue running even after the batteries have been removed.

You should nevertheless check the time and date before each series of measurements.

The time and date can be set with the analysis software. You can also set the time and date manually by holding down the Start key and then pressing the Event key. Now you are in the "Set time" mode. Use the Start key to change the number in question, and use the Event key to jump to the next.

3.4.4 Transfer of patient ID

The monitor has to be prepared with the transfer of the patient ID. This enables a clear assignment of the data to the correct patient. (Please also take a look at chapter 4.6.2 downloading)

3.4.5 Specifying the measurement protocol required

You can use the Protocol key to choose one of nine (1–9) protocols.

Once you have completed one measurement, the protocol cannot be changed until deletion of all data has been completed.

The protocols can be set using the analysis software.

You can set the protocols manually by holding down the Day/Night and then pressing the Event key. Use the Start/Stop key to change the protocol, and confirm with the Event key.

Protocol	Day time	Night time	Measurements per hour	Acoustic signal	Value shown on display
1	08:00 00:00	23:59 07:59	4 2	YES NO	YES
2	08:00 23:00	22:59 07:59	4 1	YES NO	YES
3	07:00 22:00	21:59 06:59	4 2	YES NO	NO
4	08:00 00:00	23:59 07:59	4 2	YES NO	NO
5	18:00 10:00	09:59 17:59	4 2	YES NO	YES
6	07:00 00:00	23:59 06:59	4 2	YES NO	YES
7	06:00 23:00	22:59 05:59	4 2	YES NO	NO
8	07:00 09:00	08:59 23:59	6 4	YES NO	YES
9	09:00	08:59	30	NO	YES

Table 3.0 Protocol list

Note: Protocols 1 and 2 are set as standard, but can be changed using the analysis software versions (see sections Analysis unit: Windows software, preparing the monitor). Protocol 5 is suitable for night-time activity (night shift). Protocol 9 is the Schellongtest.

3.4.6 Putting on the monitor and starting measurement

Put the equipment pouch on the patient. By varying the length of the strap you can use it either as a hip belt or a shoulder strap. Alternatively, use a normal belt that matches the patient's clothing. Now fit the cuff on the patient; correct cuff fit is extremely important for accurate measurement (see Figure 3.2).

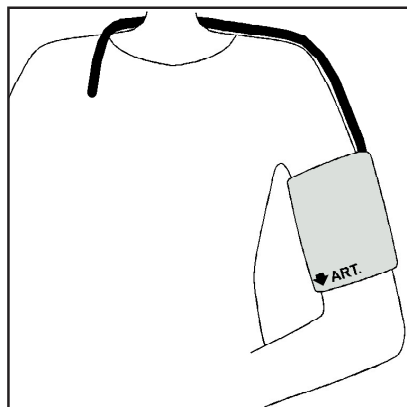


Figure 3.2 Putting on the cuff

The cuff can also be worn over the shirt or blouse. We recommend that the cuff be fitted on the bare upper arm. In this case, bring the tube out between the buttons of the shirt or blouse, around the back of the neck to the ABP recorder on the right-hand side.

Please note the following points:

IMPORTANT:

1. When putting the recorder on the patient, the recorder must not be connected to other external equipment!
2. The cuff must fit so that the air tube cannot be kinked or compressed at any point. The tube connection on the cuff should therefore be directed upwards. The air tube should allow free movement of the upper arm and should run over the back of the neck to the other side of the body.
3. It is essential to ensure that the artery symbol lies over the brachial artery. When the cuff fits correctly, the metal strip is on the outer aspect of the upper arm (elbow side)
4. The material loop must cover the skin under the metal strip.
5. The lower edge of the cuff should be about 2 cm above the bend of the patient's elbow.
6. The cuff should fit relatively snugly on the upper arm. Correct fit can be checked by a simple test: there should be room to slip one finger under the cuff.
7. The correct cuff size is also important if blood pressure is to be measured correctly. To obtain reproducible results, standardised monitoring conditions are required, i.e. the cuff must be the correct size for the patient. Using the tape measure supplied, measure the circumference at the middle of the upper arm and select the appropriate cuff:

Upper arm circumference	Cuff
20 - 24 cm	S
24 - 32 cm	M
32 - 38 cm	L
38 - 55 cm	XL

7. Connect the air tube from the cuff to the ABP recorder. Push the tube firmly over the connection until a click is heard as it engages; to disconnect, simply pull back the outer metal ring on the plug.
8. The recorder is now correctly fitted and ready for use.

Once all these steps have been completed, the monitor can be started. First, a manual measurement must be carried out by pressing the "START" key. This measurement is used to determine whether the recorder is working correctly.

If faults occur, check once more that the procedure followed in setting up and fitting the equipment was correct. Should this not solve the problem, repeat the setting-up procedure.

Only after a successful manual measurement can the patient be allowed to leave with the equipment.

Interrupting measurements

During a measurement, **ANY** key can be used to interrupt the measurement. The display then shows "-StoP-" and beeps 5 times. The process is also saved in the table of results under "Interruption".

Important: Before beginning a 24-hour monitoring session, take the patient through section 2: Operating instructions.

3.5 Technical data

Pressure measurement range:	Systolic 60 to 290 mmHg Diastolic 30 to 195 mmHg
Accuracy:	+/- 3 mmHg in the range indicated
Static pressure range:	0 to 300 mmHg
Pulse range:	30 to 240 beats per minute
Method:	oscillometric
Measurement intervals:	0, 1, 2, 4, 5, 6, 12 or 30 measurements per hour
Monitoring protocols:	2 modifiable interval groups
Storage capacity:	300 measurements
Battery capacity:	> 300 measurements
Operating temperatures:	+10°C to +40°C
Operating humidity:	15% to 90%
Storage environment: -	20°C to 50°C and 15% to 95% humidity
Dimensions:	128 x 75 x 30 mm
Weight:	approx. 240 g including batteries
Power supply:	2 NiMH batteries, each 1.2 V and at least 1500mAh (AA, Mignon) 2 alkaline 1.5 V batteries (AA, Mignon)
Interfaces:	Serial (cable) Infrared IEM-specific Bluetooth (Class 1/100 m)

3.6 Symbols

Battery symbol

Meaning: The symbol shows the type of voltage.

Disposal of used electrical and electronic appliances

Meaning: This symbol on a product or its packaging indicates that the product should not be treated as normal household waste, but must be returned to a recycling point for electrical and electronic appliances. You can obtain further information from your local authority, local waste disposal plants or specialist dealers.

Defibrillator rating

Meaning: BF defibrillator rated equipment

Caution, consult accompanying documents

Meaning: caution, consult accompanying documents

Manufacturing date

Meaning: Date of manufacture

CE mark

Meaning: The device fulfils the basic requirements of the European Directives on medical products.

4. Hypertension Management Software for Windows®

Blood pressure is measured by the ABP monitor and the reading stored. On completion of measurements, the stored readings are transferred to your computer, and you can then evaluate them according to your requirements.

A patient file is prepared, containing all important data such as:

- * Personal patient ID number (obligatory entry)
- * Name (obligatory entry)
- * Address, telephone number
- * Personal data (age, sex etc.)
- * Medication

The software program offers you various options for analysis. You can choose to display the results on your screen or print them out directly as:

- * Display of all individual measurements
- * Statistical analysis with mean blood pressure values for the whole day, and for the daytime and night-time periods and the first hour after waking, plus hourly mean values
- * Extreme values (maximum and minimum values)
- * Percentage incidence of values above a predetermined limit
- * Standard deviation (variability)
- * Calculation of day/night reduction
- * Graphs such as:
 - * Curve of measurements
 - * Bar chart of measurements
 - * Histogram of blood pressure distribution
 - * Curve comparison for optimising treatment
 - * Envelope curve of hourly mean values
 - * Curve of blood pressure changes
 - * Correlation
 - * Pie chart of percentage of values exceeding limits

This equipment allows rapid and simple visualisation of the course and fluctuations in blood pressure over the day and night. Medication can be adjusted to take the best possible account of these changes.

Using the same program, readings from the "Klock®" - the blood pressure monitor wristwatch - and the Stabil-O-Graph® - the upper arm blood pressure monitor - can also be entered and the data managed. An infrared eye is required for data transfer. For further information, please contact your specialist dealer or I.E.M. GmbH direct.

4.1 Installation

4.1.1 Installation in Windows 95 / 98 / NT / 2000 / ME / XP

Switch on your PC. Insert the CD into the CD-ROM drive. If installation does not start automatically, choose "Start", enter <cd>:\SETUP and click ENTER (<cd> represents the CD-ROM drive).

The following window will appear:



Figure 4.1.1

USB cable:

If you are using the USB cable for data transfer, you must first install the corresponding driver. At this time, the USB cable must not yet be plugged into your computer! Select the "USB cable driver" button and follow the instructions until driver installation is complete.

Hypertension Management Software:

To install the Hypertension Management Software, select the "HMS Installation" button and follow the instructions until installation is complete.

Network installation:
Contact your practice IT manager

4.1.2 Update installation

Should an older version of HMS for Windows® already be installed on your PC, proceed exactly as for the initial installation (see 4.1.1). All existing patient data and measurements will be retained.
To use the Mobil-O-Graph NG® you need Version 7.0 or later.

4.2 Program structure

The program is designed to allow you to move step by step through it from the first screen displayed. You can either use the mouse or move by using the function keys F1–F12. You can also use the program with the “Alt” key + the underlined letters and/ or the cursor keys (arrow keys). The description given below in this manual is for mouse operation only, as this is the most convenient method, but you can, of course operate without a mouse, as is clear from the instructions.

4.2.1 Menu bar

The menu bar displays all operating or selection criteria for the screen in question.

Patients Measurements Recorder Settings Window ?

Figure 4.2.1: Menu bar (excerpt)

4.2.2 Toolbar

Using the toolbar (see Figure 9.1: Menu bar (excerpt)) you can simplify and speed up your work, as the main functions (buttons shown on the active areas) are directly available by clicking the appropriate symbol.
If you do not wish to use the toolbar, you can remove it from the screen, by going to the “Settings” menu and clicking on the “Toolbar” command. Click this command again to make the toolbar visible again.

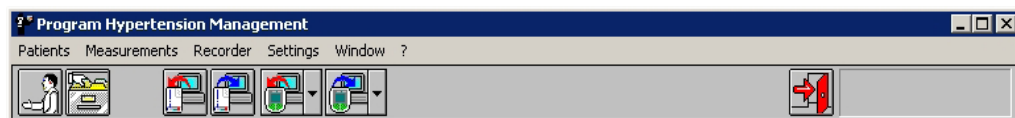


Figure 4.2.2: Toolbar

4.3 Brief instructions

- 1) Load existing patient or set up a new patient.
Note: After setting up a new patient, selection of the available series of measurements must be discontinued, as no monitoring session has yet been recorded.
- 2) Connecting to the recorder (co mode). See section Preparing the monitor.
- 3) Send patient ID number to recorder, and check the protocol used and time.
- 4) Close HMS for WINDOWS and disconnect the monitor from the computer.
- 5) Switch recorder off and on again. When the time is displayed, the recorder is ready to be fitted to the patient as described in the section Preparing for monitoring.
- 6) After a 24-hour monitoring is complete, connect the recorder to the PC and download the readings from the recorder as described in the section Analysis

4.4 The first steps

Open the ABPM program via START – PROGRAM HYPERTENSION MANAGEMENT. Using the mouse, now click on the function you require on the menu bar. The submenu will then appear and you can start the function by clicking the appropriate command. The process is even quicker if there is a symbol on the toolbar for the function you require. Then simply click the appropriate symbol. The number of symbols displayed depends on the screen resolution you have set. To allow you to get to know the program, a fictitious patient has been set up as an example.

4.4.1 Example

Loading:

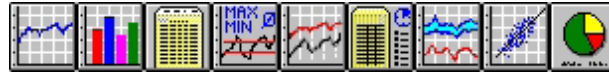


You can look at the data for the sample patient, John Doe, by clicking “Patient” on the menu bar and selecting the command “Load”. Alternatively, you can click on the symbol on the left. On the screen that now appears, mark the patient John Doe, either by entering the name in the “Find name” field or directly clicking on the name “Doe” in the Table, so that the name is highlighted in colour. Then click “OK”. The window with the patient’s data will be built up.

Measurements:

If you now click “OK” again, the “Existing measurements” window will appear. You can now select a series of readings by marking it and clicking “OK”. You then return to the basic screen, although you will notice that you now have more choices of symbols than before and you can also open a larger number of commands on the “Measurements” menu.

Evaluations: Now simply run through the various evaluations by clicking on the symbols..



This will give you an initial overview of the variety of analysis and display options offered by the program.

You will always return to the basic screen when you click Close.

4.4.2 Help



You can call up various Help topics from all parts of the program. It is also possible, for example, to call up the Direct help screen, which gives explanations on the topic you are currently processing.

In addition, however, you can also have displayed detailed help screens on particular questions or topics. To make use of this option, call up the menu. Under the menu commands "Contents", "Search" and "Use help", you have a wide variety of question options, topic overviews and explanations of functions.

4.4.3 Window

Using the "Window" menu you can see how many windows you have opened at one time. Click the appropriate window to move backwards or forwards between the various analyses.

4.4.4 Closing the program



To leave the program click "Close" either using the button or the command in the "Patients" menu.

4.5 Patient data

4.5.1 Entering new patients in the database



Using the button or the "New" command in the "Patients" menu, call up the input screen for entering new patients

Figure 4.5.1: Patient data

Move through the record sheet during input using the tabulator key. For later allocation of measurements, however, the only other important feature apart from the ID number is the name (both inputs are sort and search criteria). All other information is voluntary. Using the "Diagnosis" field, you can enter your diagnosis for the patient in a text field. The "Extended" switch opens a box for entering more detailed patient information, such as medication, telephone number, email address or the one-off blood pressure result taken manually. Once entered, the values are available at all times and can also be changed.

To save the new patient, click OK

If you do not wish to save the data entered, click Close.

You will again be asked whether the data should be saved. If you click "No", the data will not be saved and the window will be closed.

4.5.2 Loading stored patient data



From this list, select a patient, to see their findings to date, to program the recorder for this patient or to allocate a new series of results to the patient.

From this list, select a patient, to see their findings to date, to program the recorder for this patient or to allocate a new series of results to the patient.

Surname	First name	Patient-ID	Date of birth
Zweibein	Ferdinand	999999999	02.08.45

Figure 4.5.2: List of patients

You can load the patient either by double-clicking his or her name in the list, or by marking the name with a single click and then opening using the "OK" button.

Search: Enter the surname of the patient in the "Name search" field to find him or her quickly. Alternatively, you can enter the patient's ID number in the "Pat. no. search" field.

List: If you click on the "Print list" field, the whole list of patients will be printed, sorted either alphabetically by name or by patient ID numbers, depending on the sort criterion selected.

4.5.4 Exporting readings (ANSI file)

It is possible to store the reading from long term monitoring, which have been loaded from the recorder and then evaluated, as a report of readings, so that they are available to your practice system.

Set up: If, after selecting a series of measurements for a patient in the "Measurements" menu, you click the command "Save values in ANSI format", the program automatically converts the data into an ANSI file. You can then access these data from your practice system.

Export file: If you wish to change the specified name of the export file, you can do so via the "Settings" menu with the command "ANSI export file". In this dialogue box you can enter the required name and directory of the export file.

4.6 Mobil-O-Graph NG® blood pressure monitor

4.6.1 Preparing the monitor

Connection
via cable:

Make sure that your PC is switched off. Connect the interface cable supplied to the selected interface of your PC. The other end of the cable is now connected to the recorder, which must be switched off. The plug is inserted in the port on the recorder until it engages. Please do not force it! Switch on the monitor. The letters "co" should be visible on the display. If your PC does not have a free serial port, you can also use a USB port with the USB combi cables.

Connection
via infrared:

Switch on the Mobil-O-Graph NG®. Open the interface menu by pressing and holding the Start key, and then pressing the Day/Night key. You can now choose between the flashing messages "co", "bt" and "PAIr" with the Start key. Select "co" and confirm this with the Event key. The letters "co" now longer flash and the interface is active. Now point the "IR-Med" IR interface, which is connected to your PC, towards the IR interface of the Mobil-O-Graph NG® at a distance of 10–15 cm. Two IR interfaces are available from I.E.M. (see also Figure 4.7.1):

- IR-Med with serial interface to PC
- IR-Med with USB interface to PC

Note: - The devices are not compatible with existing built-in infrared interfaces, for example in laptops.
 - If there is no activity, the "co" mode will stop after 10 minutes.

Set-up:

You must first decide on a free serial port on your PC. In the "Settings" menu you can use the "Interface for monitor" command to open a dialogue window, in which you can select the serial port you wish to use to connect the Blood pressure monitor to your PC. (Note: COM1 is generally already being used for the mouse.) Click "OK" to save your setting.

Preparation:



Using the "Monitor" menu, the "Prepare" command will take you into the window for programming the recorder for the next monitoring session. You can also reach this window via the button shown on the left.

You now select the measurement protocol to be used for the monitoring session. The parameters you will enter include the time intervals (throughout the day) at which measurements are to be taken (intervals always start on the hour), and whether the values measured are to be shown on the monitor display after a measurement or not. These changes can be undertaken in protocol 1 and protocol 2.

Important:

You must in any case enter all the intervals, even if you wish to select only one interval for the whole of the monitoring session. In this case please set the same characteristics of measurement frequency and acoustic signals for all intervals; only the start times must be different.

Acoustic signal:

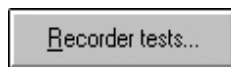
Using the "Buzzer" option, choose whether or not an acoustic signal is to be emitted to warn of a measurement.

Monitor clock:

Using the "Set recorder clock" function, you also have the option of setting the time shown on the monitor to that of your computer.

Monitor test:

To ensure that the monitor is functioning correctly, you can test its functions before starting the monitoring session. Click the button



to get to Figure 4.6.1:

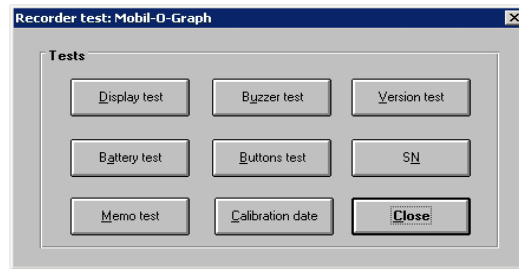


Figure 4.6.1: Monitor test

Click the individual test functions to test the equipment accordingly.

Patient ID number: The patient ID number just stored in the monitor is saved by clicking the active area, so that readings can later be allocated to the patient automatically when downloaded to the PC.

Send patient number: 000009999999999

Switching off: Now switch the recorder off, and disconnect the PC connector cable. Now switch the equipment on again; after the battery voltage has been displayed, the display will show the time.

Protocol: By pressing and holding the Day/Night key while pressing the Event key, check whether the right measurement protocol has been saved (protocols 1 and 2 can be changed as you wish). The protocol can also be selected directly from the PC in "Preparing the monitor", by marking the button with an X by clicking.

☒ Activate automatically when sending

Note: It is extremely important always to check the protocol set before starting the long-term monitoring process, since no further change can be made to the protocol once the first measurement is under way.

Measurement: Please follow the instructions in the sections Preparing for monitoring, section. 3.4/ Putting on the Monitor and starting measurements section. 3.4.6 and Instructions for the patient section.2.6.

4.6.2 Downloading from the monitor after the 24-hour monitoring session



Reconnect the monitor to your PC, and the switch the monitor on. "co" is shown again on the display. If you are using the IR interface, select "co" using the keys of the Mobil-O-Graph NG® (see section 4.6.1). If you stored the patient's ID number in the recorder during preparation of long term monitoring, you now only need to select the "Download" command from the basic screen in the "Monitor" menu or click the button, to have the measurements allocated automatically to the patient. However, if you have not stored the patient ID number in the recorder, you must first load the patient (see section 4.5.2), in order to use the "Download" command in the "Monitor" menu.

After downloading you will be asked whether you wish to delete the data in the recorder. If the transfer was completed without an error message, click "Yes" to allow you to prepare the recorder for the next patient. If you do not delete the data at this point, you will be reminded to do so again later when you are setting up the new patient.

4.7 “Stabil-O-Graph®” (optional)

4.7.1 Infrared data transmission using the “Stabil-O-Graph®”

The Stabil-O-Graph® provides a wireless method of exchanging data with a computer (PC) via an infrared interface. Data exchange is possible as soon as the infrared connection between the device and the computer has been established.

I.E.M. provides two IR interfaces:

- IR-Med with serial interface to PC
- IR-Med with USB interface to PC

Note: The devices are not compatible with existing built-in infrared interfaces, for example in laptops. The IR interface provided by I.E.M. can also be used for your Mobil-O-Graph NG®.

System requirements

- * Computer with a free serial 9-pole interface, or computer with a free USB interface.
- * Successfully installed HMS software
- * Separate infrared interface (see Fig. 4.7.1)

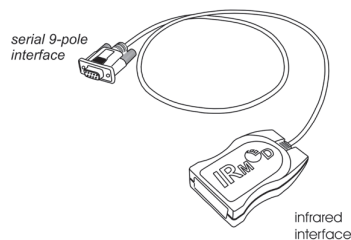


Figure 4.7.1: Infrared interface with 9-pole serial connection

Installation instructions for the serial infrared interface

1. Take care to ensure that your PC is switched off. Connect the infrared interface to a free serial port at the rear of the computer and tighten the screws.

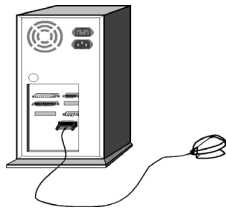


Figure 4.7.2: Connecting infrared interface to the computer

2. Switch on the computer and start the software.
3. Set the serial interface in the software, by choosing “Interface for monitor” in the “Settings” menu.

Installations instructions for the USB infrared interface

1. **First**, install the **USB driver** found on the CD of HMS software before connecting the IR interface to the PC for the first time.
2. Set the IR interface in the software by choosing “Interface for monitor” in the “Settings” menu.

4.7.2 Preparing for monitoring

- Ensure that the batteries in the unit have enough capacity, i.e. date and time are shown, but the battery symbol is invisible.
- Switch the unit into transmission mode. First press and hold the START key, then also press the MENU key. “co” will appear and the unit is ready to exchange data.
- Place the unit and the infrared interface on a smooth surface, about 2–10 cm apart. Please point the infrared interface towards the side window of the unit.
- Take care to ensure that the unit remains in the same position during the whole transmission.

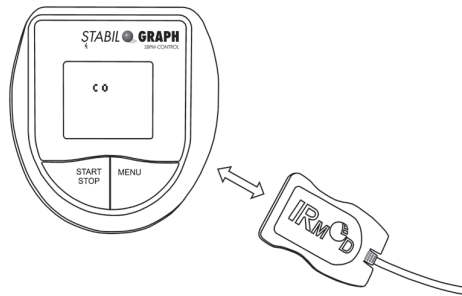


Figure 4.7.3: Pointing the infrared interface

- Start your Hypertension Management software for Windows..
- Either create a new patient using the  function, or use the



function to select an existing patient; confirm with "OK".

- Click the  function to prepare the unit.

- In the window that opens

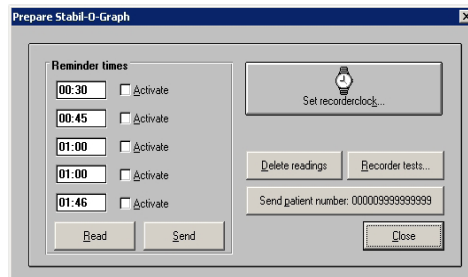
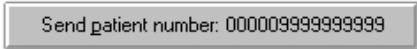


Figure 4.7.4

you can enter, activate and send reminder times, perform various tests, and set the current time.

- Now click the  field
- Click  to close the program.
- After data transmission is complete, please pick up the monitor and press any key (time and date are shown).
- Now the device is ready to carry out blood pressure monitoring again.

4.8 Analysis

After you have downloaded the readings into the HMS program, you have many analysis and representation possibilities to choose from. These are described below.

Important: To enter the basic screen for the individual representations, you must first call up the personal data for the patient you require via the "Patient" menu using the "Load" command. After you have clicked "OK", you can decide on particular measuring series by marking it and clicking "OK", or double-clicking the measuring series. Now you are taken to the modified basic screen by displaying the various analyses for the selected measuring series.

4.8.1 Profile display

The Profile display shows the following values from the selected measuring series graphically as a function of time in a diagram:

- * systolic values
- * mean values
- * diastolic values
- * pulse

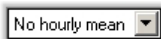


You can call up the Profile display either via the "Measurements" menu using the "Profile" command, or by clicking the button. A diagram is drawn, from which you can read off the following details:

Blood pressure: On the left edge of the screen you can see the ordinate for blood pressure in mmHg units. You can move the axis up and down using the scroll bar. The systolic, diastolic and mean blood pressures are related to this left-hand axis.



Use the check box on the lower edge of the screen to switch the display of mean values on and off.



Use the next selection field at the lower edge of the screen to select whether, and which hourly mean values to show additionally (up to a mean over 8 hours).

Pulse:



On the right edge of the screen is the ordinate scale for pulse. The unit is 1/min. Here, too, you can use the scroll bar to shift the axis. Pulse readings are related to this right-hand axis.

Use the check box on the lower left edge of the screen to switch the pulse curve on and off.

Time: The lower axis of the diagram shows time in hours. The bars above it show the intervals for day (red) and night (blue). Day counts all three daytime intervals together. "M" signifies manual measurements.

Curve colours: The area enclosed by the curves for systolic and diastolic values is filled in with a colour that contrasts with the background. You can change the curve and background colours as you wish, by clicking the Colours button at the bottom right edge of the screen. In the window that now appears you can change the colours by clicking the colours button for the curve in question, select a colour and confirm your selection with "OK".

Limits: The upper limits set for the individual values for single measurements and mean values are entered as a theoretical curve (dotted line). If you wish to change the values for these blood pressure limits, use the "Limit" command in the "Settings" menu.

Individual values: You can have the exact values for a measurement displayed by clicking any point on the blue screen at the time of the measurement. A vertical bar and a small window with the exact details of time and readings are then displayed.

If you now sweep the mouse over the graph, the display will be continuously updated in the small Info window. You can move this window to a different position by clicking any point within it and dragging it with the mouse. By clicking on the screen background you can close the Info window again.

Comments: By clicking the Comments button you will open a window in which you can make entries relating to the series of measurements, which will then also be printed out with the other data.

Printout:



If you wish to print out the profile display, please click on the appropriate button or the "Print" command in the "Patients" menu. Selecting the "Page 1 (Blood pressure profile, Notes)" criterion to print out the profile display. Using the Printer setup button you can select and install your printer.

Please note that the last axis setting (set with the scroll bar) will be used for the printout.

Close: Close the profile display chart by clicking Close.

4.8.2 Histograms



The histogram displays the frequency distributions of systolic and diastolic values and pulse in the form of three bar charts.

You can call up these charts from the "Measurements" menu using the "Histogram" command, or by clicking the button.

This chart displays the percentage within each decile occupied. In other words: the percentages of values measured for systolic and diastolic blood pressure, in the ranges, for example, 80–89 mmHg, 100–109 mmHg etc., are calculated. Similarly for pulse, the frequencies of values in the ranges 50–59 1/min, 60–69 1/min etc., are calculated. These percentages are displayed as a bar chart. For example, the bar over the number 130 for systolic values shows the frequency of all systoles in the 130s, that is, all values in the range 130 to 139 mmHg.

Basis: Using the selection field, choose a period to be used as the basis for calculations.

Range: "Total" means that all measurements are to be included; with "day" only measurements taken during the three daytime intervals will be considered, similarly, with "night" only values during the night interval. Should no data be available for any interval, you will be informed accordingly.

%-Skala: Using the button, set the maximum frequencies that can be displayed. The lower the maximum values you choose, the higher the resolution you will obtain. Should frequencies occur that are higher than the maximum set, the bars concerned will be displayed in dark green and a small warning field will flash on the left next to the chart in question. In this case, choose higher maximum values for the percentage axis.

Recalculate: The "Recalculate" button is active only if (with the frequency display open at the same time) you have switched the Inclusion in statistics setting on and off for at least one measurement in the table of measurements and, on returning to the histogram you have answered No to the question "Recalculate?". This button recalculates the frequencies retrospectively, taking the changes into account.

Comments: Click the "Comments" button to open a window in which you can make entries relating to the series of measurements. These entries in the Comments field apply to all other Comments fields.

Printout:



If you wish to print out the frequency distributions, please click the button, or on the "Print" command in the "Patients" menu. By selecting "Page 2 (Frequency distributions)" with the appropriate base you can print out the histogram.

Please note that the last % scale prepared will be used for the printout.

Close: Close the bar chart or histogram by clicking the "Close" button.

4.8.3 Table of measurements



Using the "Measurements" menu you can have the readings displayed in tabulated form by clicking the button for the monitoring session loaded, or by using the "Table of measurements" command.

Comments: To store a "comment" on a particular reading, click in the last column of the relevant row. Enter the comment and press the Enter key.

Marking: You also have the facility to mark measurements and so exclude them from the statistical analyses.

This is useful if a measurement is a complete outlier and would therefore falsify a representative long-term evaluation.

Click in the first column of the relevant row. The number of the measurement is now in brackets and will from now on be excluded from any statistical assessment. Click the number again to remove the marking.

Limits: If particular values lie outside the limits set, this will be indicated by an asterisk before and after the value (e.g. *50*). In the printout a symbol will appear with a raised index finger, designating a value over the limit.

Comments: Click the "Comments" button to open a window in which you can make entries relating to the series of measurements. ANSI Export contains all jobs from the Table of measurements.

Ausdruck:



If you would like to print out the Table of measurements, please click the button or on the "Print" command in the "Patients" menu. Selecting "Page 3 (Table of measurements)" to print out the tabulated list.

Close: Close the Table of measurements chart by clicking the "Close" button.

4.8.4 Report of findings (statistics)



The Report of findings can provide various statistical statements on systolic and diastolic blood pressure.

Open this window by clicking the "Report of findings" command in the "Measurements" menu, or just by clicking the button.

Evaluations for day and night are performed separately.

Intervalle:

The Day and Night displays show the mean values and scatter (empirical standard deviation) for blood pressure values.

How many are	than	Result:
Greater/equal	140	27 of 33

The List field is used to calculate the number of readings lying above a certain limit. The individual values from the "Global limits" dialogue field are used as standard. The result shows how many values from all valid measurements satisfy the comparison; the percentage is shown by the bars underneath.

If no data are available for an interval, the grid remains empty. This may be the case if no measurements were taken during the period in question, or if all measurements have been excluded from statistical evaluation (by corresponding marking in the Table of measurements).

Reduction:

This field shows the percentage reduction in mean blood pressure values (= mean values) between the day and night intervals, and the mean blood pressure value within one hour of starting the daytime interval.

Day/Night decrease, early morning		
Decrease	15,0% / 20,0%	---> Non-Dipper
Morning average	149 / 99 mmHg, 82 /min	

Recalculate:

The Recalculate button is active only if (with the frequency display open at the same time) you have switched the Inclusion in statistics setting on and off for at least one measurement in the table of measurements and, on returning to the histogram you have answered No to the question "Recalculate?". This button recalculates the frequencies retrospectively, taking the changes into account.

Comments:

Click the "Comments" button to open a window in which you can make entries relating to the series of measurements. Note: the BDT Export file contains all the information from the Report of findings.

Printout:



If you wish to print out the Report of finding, please click the button, or on the "Print" command in the "Patients" menu.

Select "Page 4 (Report of findings)" to print out a statistical analysis of the measurement series

Close:

Close the "Report of findings" statistical analysis by clicking the "Close" button.

4.8.5 Comparing two series of measurements



Prerequisite: at least 2 measurement series must have been recorded for a patient. This evaluation allows you to display the systolic, mean and diastolic blood pressure values and pulse for the previously downloaded main series of measurements and a further comparative series in one bar chart. Call up this display by entering the command "Comparison of two series of measurements" in the "Measurements" menu, or simply click the button.

Selection:

When you call up this display, a selection window will appear in which you can select the reference series of measurements by clicking. The main series selected previously will be displayed in the top part of the screen. Confirm with "OK", to construct the chart showing both series of measurements.

Blood pressure:

On the left-hand edge of the screen you will see the ordinate markings for blood pressure values in mmHg. The values for systolic, diastolic and mean values (blood pressure values) are related to the left axis of co-ordinates.



Use the check box on the lower edge of the screen to switch the display of mean values on and off.

Puls:



On the right edge of the screen is the ordinate scale for pulse. The unit is 1/min. Use the check box on the lower left edge of the screen to switch the pulse curve on and off.

Comments: Click the "Comments" button to open a window in which you can make entries relating to the main series of measurements.

Close: Close the comparison of two series of measurements by clicking the "Close" button.

4.8.6 Hourly mean values

It is also possible to display in tabulated form the hourly mean values for blood pressure and pulse rate from the series of measurements selected.

To do this, open the tabulated display from the "Measurements" menu using the "*Hourly mean values*" command.

Selection: You can now also select which hourly mean values you wish to display. You can select hourly values up to a mean value over 8 hours.

Comments: Click the "Comments" button to open a window in which you can make entries relating to the series of measurements.

Close: Close the tabulated display of hourly mean values by clicking the "Close" button.

4.8.7 Changes in blood pressure

Use the "*Measurements*"- "*Change in blood pressure*" menu to monitor your patient's morning rise in blood pressure.

4.8.8 Correlation

Activate the "*Measurements*"- "*Correlation*" menu option, to open a window showing a correlation of diastolic and systolic blood pressure. Each point on the graph corresponds to a measurement.

From the Total combination box you can select whether all readings, only daytime measurements or only night measurements, should be included in the correlation. In the "*day*" and "*night*" settings, the limits set are also displayed (see also 4.9 Settings, section Limits).

4.8.9 Comments

As already described in previous sections, you can enter any notes and comments on a particular series of measurements for a selected patient in a special window. As well as being shown when viewing an evaluation by activating the appropriate button, these comments can also be called up by clicking the "*Comments*" command from the "*Measurements*" menu.

For each monitoring session for a patient there are specific entries, relating for example to medication or a Report of findings.

In this window you can enter as many lines as you wish, but only the first 20 lines will be printed (from the "*Patients*" menu using the "*Print*" command and selecting Printer output "*page 1*").

Press the "*Save*" key to store the entries and close the input window.

4.9 Settings

You have the option of establishing various settings as standard for the program:

Limit: In the "*Settings*" menu, the "*Limits*" command will take you to an input window in which limits for evaluations can be set. Pre-set values are, for example, 140 mmHg as systolic and 90 mmHg as diastolic value for the individual daytime values. If the limits are exceeded, the readings will be marked accordingly in the evaluations. You can change the limits by entering new values and confirming the input using the "*OK*" key.

Colours: Call up the "*Settings*" menu and click the "*Colours*" command to change the colours of the curves and chart backgrounds for on-screen display to suit your preferences.

Headers/Footers: You can enter an individual standard text for printout of the individual evaluations and for the top line of the on-screen display. To do so, in the "*Settings*" menu please click on the "*Headers and Footers*" command and enter the text you require (e.g. your address and telephone number) in the appropriate fields. The input will be stored when you confirm using the "*OK*" key.

5. Care and maintenance

To ensure the lasting optimum functioning of your Mobil-O-Graph NG[®], regular care and maintenance of the unit is necessary.

Cleaning and disinfection:

The user (doctor) decides, whether and when disinfection of the cuff sleeve is necessary for reasons of hygiene (e.g. after each use).

For disinfecting the cuff sleeve, I.E.M. has successfully tested and recommends the following solutions:

1. Terralin Liquid (manufacturer: Schülke & Mayr)
2. Promanum N (manufacturer: B. Braun)

Important: It is vital to observe the manufacturer's instructions for the use of these products. Always leave the solution to dry completely.

Avoid disinfecting the cuff bladder and connected rubber tube. Before disinfecting, take the bladder and the tube carefully out of the cuff sleeve. The bladder and tube may be damaged by disinfectants. Do not place the cuff sleeve in the disinfectant solution. This might damage it. When using other disinfectants not recommended by IEM, the user is responsible for proving harmless application. Never use disinfectants that leave a residue on the product or which are unsuitable for use in contact with skin.

Caution: Some patients have intolerances (e.g. allergies) to disinfectants or their components. If you have such a patient or you are not sure, remove possible residues with careful washing. The cuff sleeve can be washed in a washing machine at max. 30°C using a mild washing powder. Do not use fabric softeners or other aids (e.g. disinfectant rinses, textile deodorants). These solutions may leave residues and damage the material. The cuff sleeve is not suitable for drying in a tumble dryer.

The Velcro must always be fastened before washing.

Bladder: Wash the bladder only with lukewarm water, adding a mild cleaning solution if necessary, and wipe off. Take care to ensure that no water enters the tube opening.

Monitor: Clean the Mobil-O-Graph NG[®] only with a damp cotton cloth. Do not use corrosive liquids or solvents. Take care to ensure that no water enters the device.

IMPORTANT: If liquid does penetrate the unit, switch it off immediately and send it to I.E.M. for testing.

Pouch: Please clean the pouch only with a damp cotton cloth (do not use corrosive liquids or solvents).

Service plan:

Weekly

Check the printout carefully for:

- * Correctly entered times and intervals in accordance with the protocol
- * Times of day/night changeovers
- * Correctness of standard values (nocturnal reduction)
- * Battery voltage curve (see section Battery voltage curve)

The fully charged batteries are installed in the monitor direct from the charging unit. The battery voltage is shown in the LCD display of the ABP monitor shortly after switching on the device, for approx. 3 sec. This voltage should be at least 2.75 V to guarantee measurement over 24 hours.

Every 2 years

In accordance with the basic requirements and general Directives (e.g. 93/42/EEC, Medical Device Operators Ordinance Art. 11 and Appendix 2) the Mobil-O-Graph NG[®] monitor must be subject to metrological checks every two years.

We offer to provide you with metrological check and complete servicing. This service comprises:

- * Calibration
- * Software update
- * Functional testing: electronics, pump and pneumatic circuit.

Batteries: Rechargeable batteries age. Batteries that are damaged, or with which 24-hour monitoring is no longer possible, must be replaced immediately. The batteries will last substantially longer if they drain completely before charging them.
The batteries supplied are of the nickel-metal hydride (NiMH) type. Self-discharge of these batteries depends on the ambient temperature. It is therefore recommended to leave the batteries in the charger until they are actually needed.
We recommend using I.E.M. batteries exclusively, as their capacity and quality has been tested. Care should always be taken that the batteries' capacity is above 1500 mAh.

Battery charger: Follow the operating instructions included with the charger.

6. Troubleshooting

6.1 Principal sources of error

The following may cause faulty readings or undesired results:

- Arm movement by the patient during the measurement
- Switching off the device (e.g. at night)
- Incorrect cuff size fitted
- Slipping of the cuff in use
- Absence of successful manual measurement at the surgery
- Failure to take medication
- Incorrect protocol set by the user
- Batteries not fully or correctly charged, or worn out
- Defect in the charging unit

6.2 Battery voltage curve

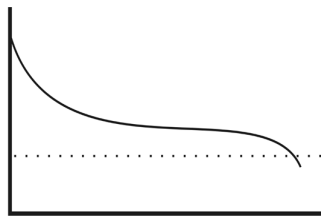


Figure 6.2.1
Rechargeable batteries OK / Batteries OK
or wrong batteries

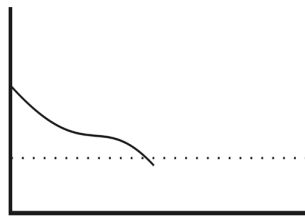


Figure 6.2.2
Rechargeable batteries old or too little capacity

6.3 Transmission errors

To avoid errors in data transfer, the monitor checks the data transmitted.

If an error has occurred, E004 appears on the display

6.4 Check list

Please run through the following check list if errors occur when using the Mobil-O-Graph NG®. Many faults are easily remedied.

- Check whether all cables are securely connected, and the infrared interface is correctly pointed towards the monitor.
- Check whether the monitor, computer and printer are switched on (depending on version obtained).
- Check whether the connector cable is fitted to the correct serial interface (COM1 to COM4).
- Check whether the batteries are adequately charged.

Note:

Some errors are associated with a permanent alarm for safety reasons. The permanent alarm can be stopped by pressing any key. If there is pressure remaining in the cuff, open the cuff immediately.

6.5 Error tables

6.5.1 Troubleshooting, Mobil-O-Graph NG®

Error symptom	Possible cause	Remedy
After replacing batteries, the clock still shows 00:00 and the date 01.01.2006	The internal buffer battery is empty. The biennial calibration has not been performed. The buffer battery is changed at the metrological check	The date and time can be set after each change of batteries. Send the device for calibration to your dealer or directly to IEM GmbH.
The measurement data can no longer be called up or displayed.	An error has occurred during saving of patient data.	Delete the patient concerned (menu bar) and re-enter.
The connection from the Mobil-O-Graph NG® to the PC is faulty.	1. The incorrect COM interface is set. 2. The cable plug or port is defective. 3. The Mobil-O-Graph® is not in transfer mode (the time is shown on the display).	1. Set the correct interface in the utilities program. 2. Examine the plug and socket on the Mobil-O-Graph NG®. Make sure that the pins are straight so that contact is assured. 3. Switch the Mobil-O-Graph NG® off and then on again, without removing the connector cable.
The patient ID number is missing.	The monitor is not initialised, i.e. the patient ID number has not been transferred during preparation for a 24-hour monitoring session.	The patient ID number can still be transferred after the measurement. The data obtained will not be affected.
No measurements were carried out during the night phase.	1. The batteries were drained prematurely. 2. The patient switched off the Mobil-O-Graph®.	1. The rechargeable batteries may be defective (please contact your dealer). 2. Alert the patient to the urgency of obtaining a complete 24-hour recording.
The message "co" does not appear on the display.	You are not in transfer mode.	1. Communication via cable: switch the Mobil-O-Graph NG® off and then on again, without pulling out the plug. 2. Communication via IR: press and hold Start again, and press Day/ Night. Use Start to select "co".
No automatic measurements are performed.	1. No manual measurement was performed after putting on the equipment. 2. An incorrect protocol was set.	1. After putting on the equipment a successful measurement must always be performed manually. Read about this in the section Putting on the monitor. 2. Set Protocol 1 or 2, as described in the section Preparing the monitor.
The measurement interval is not as you expected.	1. An incorrect protocol was set. 2. No manual measurement was performed after putting on the equipment.	1. The programmed protocol is not the protocol set in the Mobil-O-Graph NG®. Check the protocol manually (see 3.4.5) on the device. 2. Perform a manual measurement to activate the set protocol.

Error symptom	Possible cause	Remedy
Err 1	1. The patient has severe arrhythmia 2. Not enough valid heartbeats detected	1. Monitor not applicable 2. Re-fit the cuff
Err 2	1. The arm was moved during the measurement 2. Cuff not correctly fitted on the arm	1. Keep the arm still during measurement. 2. Check the fit of the cuff and the unit
Err 3	1. Battery voltage too low 2. Blood pressure outside measurement range	1. Replace the batteries 2. If this message continues, the monitor is not suitable for the patient
Err 4	1. Data transfer cable not correctly inserted in the recorder 2. Pins in plug of data transfer cable have suffered mechanical damage. 3. Measurement value not correctly transferred.	1. Plug the cable into the monitor correctly (see section Preparing the monitor) 2. Check the plug to see whether pins inside are damaged. If so, call your dealer or IEM. 3. Start the transfer again.
Err 5 bAtt	1. Battery voltage too low 2. Batteries defective 3. Battery contacts are corroded	1. Replace the batteries 2. The battery voltage is correct, but during cuff inflation, bAtt is shown on display. Replace the batteries. 3. Clean the battery contacts with a cotton cloth and some alcohol
Err 6 + possible continuous alarm until a key is pressed	1. Air tube blocked 2. Battery voltage too low 3. Blood pressure cuff is not correctly connected 4. Leak in cuff or connecting tube	1. Check the cuff for a blockage or kink in the tube. If there is a kink in the cuff tube, release the tube. Otherwise send the equipment for repair without delay. 2. Replace the batteries 3. Connect the cuff with the unit as described in section 3.3.5 4. If necessary, replace the cuff or connecting tube.
Err 7	The recorder memory is full (maximum 300 measurements and Events may be stored)	Look in the Preparing the monitor section to see how to delete data (make sure that the data have been saved)
Err 8	No readings stored	Perform a manual measurement

Error symptom	Possible cause	Remedy
Err 9 + possible continuous alarm until a key is pressed	1. Pressure remains in the cuff 2. Zero-point adjustment could not be performed successfully.	1. Wait for the cuff to deflate completely. 2. Send the unit without delay to your dealer or direct to IEM GmbH for checking.
Err 10 + continuous alarm until a key is pressed	1. Serious fault due to pressure buildup outside the measurement process (pump has switched on illegally) 2. These error messages all indicate a serious error in the program code.	Send the unit without delay to your dealer or direct to IEM GmbH for checking and repair
The analysis unit does not react to data transmission, but co is shown in the display	1. Data transfer cable is not plugged into PC 2. see also Err 4	1. Check whether the 9-pole plug of the data transfer cable is sitting securely in the interface port 2. see also Err 4
The Mobil-O-Graph NG® takes measurements every two minutes	Protocol 9 has been set in the Mobil-O-Graph NG®	Set Protocol 1 or 2, as described in section 3.4.5 Specifying the measuring protocol required.
The required protocol cannot be set by pressing the protocol key.	Measurements from the last patient are still in the memory	Delete data from the recorder, but make sure that the data have been saved (see section 3.4.2 Clearing the memory)
The monitor cannot be switched on	1. The batteries were installed incorrectly 2. The battery voltage is too low 3. Display defective	1. Install both batteries again and check for correct polarity 2. Replace the batteries 3. Send the unit for repair to your specialist dealer or direct to IEM GmbH.
A fault occurs during the first measurement.	The cuff does not fit the upper arm of the patient	Using the tape measure supplied, measure the arm circumference and compare it with the printing on the cuff.

6.5.2 Communication errors, Stabil-O-Graph®

Error symptom	Remedy
Data transfer not possible	1. The Stabil-O-Graph® must display co 2. Connect the infrared interface securely with the PC 3. Set the correct serial interface (see HMS software "Settings" or "Utility program") 4. Correct the direction, as described in section 4.7 5. There are no readings stored in the Stabil-O-Graph®. The readings are saved only after 10 seconds. Before this is complete, do not touch any Stabil-O-Graph® key. 6. The computer's serial interface is defective.
Measurement starts instead of displaying co	Press and hold START/STOP until all segments have been displayed, then additionally press MENU until co appears.

7 Charging units

A common reason for reduced battery life is incorrect charging. If these batteries are used incorrectly they lose capacity and are no longer able to supply the Mobil-O-Graph NG® with power for 24 hours. Should the batteries have lost their charge during storage, please recharge accordingly. For further information, please see the notes in the Operating Instructions for the charging unit.

Tips:

- Charge these batteries fully before using them for the first time.
- Please note that these NiMH rechargeable batteries reach full charging capacity only after their 4th charging cycle.
- Recharge the batteries if they have been unused for some time.
- Do not use batteries that have been stored at above 45°C or below 0°C.
- To protect your batteries, avoid deep discharging.
- Should the service life of the batteries sink significantly, replace all the batteries simultaneously. Never use old, used batteries together with new, unused ones!

Care and maintenance:

- Do not immerse the unit in water.
- Clean the unit only when it is separated from the power supply.
- Clean the unit with a soft, dry cloth.
- Do not use cleaning fluids

Safety instructions:

- Never load non-rechargeable batteries. These may leak or explode and cause damage to health.
- Do not dismantle the unit or the charger cable.
- Cease using the unit if it has been damaged.

7.1 M2X

If you are using the M2X or a structurally identical charging unit, you must observe the following instructions:

- Before using it for the first time, plug the power plug supplied into a mains socket and the small plug into the jack on the charging unit.
- Open the cover of the unit and place 1 to 4 rechargeable batteries individually into the upper level of the charger. (Check their polarity!) Each shaft has a separate charging circuit, and each of these four circuits can charge a battery.



→ LED red light: Battery is being charged (quick charge)
LED green light: Battery is charged (trickle charge)

Bild 7.1 M2X model

If the red LED light does not come on, check whether the battery is placed correctly in the shaft and is touching the contacts. Should the red light still not come on, the battery can no longer be charged. If the red light does not change to green after more than two hours of charging (for batteries up to 2000 mAh), the battery is no longer capable of being fully charged and should be replaced.

If you leave a battery in the unit, it will be trickle charged so that the battery does not discharge. If you do not use the unit for some time, remove the plug from the power supply.

Because of the unit's high charging current (950 mA for AA and 500 mA for AAA), it should be used only for high-quality rechargeable batteries >1000 mAh (AA) and >500 mAh (AAA). The M2X quick charger has four charging circuits, which enable 1, 2, 3 or 4 batteries to be charged simultaneously. Mignon (AA) micro (AAA), NiCd (nickel cadmium) and NiMH (nickel metal hydride) batteries can be charged. The unit has overload protection.

7.2 TL 4+4 Ultrafast Charger

If you are using the TL 4+4 Ultrafast Charger or a structurally identical charging unit, you must observe the following instructions:

- Before using the unit for the first time, remove the coloured adhesive strip beside the magazine (if available) and connect the power cable with the charging unit and a mains socket.
- Place 1 to 4 rechargeable batteries individually into the upper charger. (Check their polarity!)
- Note the signals of the upper red LEDs. They indicate the charging process and the completion of charging.
- Remove the charged batteries. (The red light in question must be flashing!)



→ Charging station

→ LED red light: Batteries are being charged (quick charge)
LED flashing red light: Batteries are charged (trickle charge)

Note: when battery is correctly loaded, the light will flash once per second. Fast flashing (5 times a second) indicates a faulty rechargeable battery or a non-rechargeable battery.

→ Dock (prevents self-discharge)

Bild 7.2 TL 4+4 Ultrafast Charger model

You can find further instructions in the operating instructions.

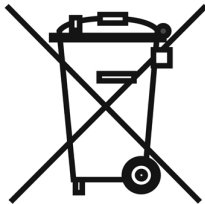
7.3 Operation with alkaline batteries

The Mobil-O-Graph NG[®] can also be operated with alkaline manganese batteries. These batteries are not rechargeable and must therefore not be placed in the charging unit.

Two alkaline batteries at full capacity have a higher voltage (min. 3.1 V) than 2 NiMH batteries (min. 2.75 V). The battery indicator on the recorder will therefore also show a higher voltage.

Place these batteries in the battery compartment in the same way as the rechargeables (check polarity). After 24 hours remove the batteries and dispose of them.

Remember that batteries are special waste and must be disposed of separately from domestic refuse.



8. Guarantee and repair conditions

Guarantee:	I.E.M. provides a two-year guarantee on the hardware, that is, the actual ABP monitor. The guarantee becomes invalid if the unit has been opened, incorrectly used or damaged deliberately or as a result of negligence. The guarantee also becomes invalid if the equipment has been repaired by an unauthorised person. The user may change only the NiMH batteries. The guarantee does not cover normal wear and tear of rechargeable batteries, connector cable, cuffs and tube. Under the terms of the guarantee, I.E.M. is liable only for faults that existed when the customer acquired the product (and which may have come to light only later). Furthermore, this guarantee is not applicable if the fault has been caused by failure to observe the operating instructions, incorrect use, liquids, damp, extreme temperature and climate conditions or fluctuations, or by corrosion, oxidation, unauthorised interventions or attempts to connect it, unauthorised opening or repair, attempts to repair using unauthorised parts, incorrect operation, incorrect installation, accidents, acts of God, spillage of food or drinks, chemical effects or other external influences (faults in wearing parts such as batteries, which have only a limited lifespan, are also excluded), unless the fault is directly due to flaws in material, construction or manufacture.
Repair:	If the unit has functional errors, or if constant measurement anomalies occur, you should contact your Mobil-O-Graph NG® specialist dealer or I.E.M. GmbH direct for repair and despatch instructions. Caution: Do not open the housing. If the unit has been opened, all guarantees become invalid
Metrological checks:	In accordance with the basic requirements and general Directives (e.g. 93/42/EEC, Medical Device Operators Ordinance Art. 11 and Appendix 2) the Mobil-O-Graph NG® monitor must be subject to metrological checks every two years.
Liability clause:	In all cases in which I.E.M. is obliged by contractual or legal reasons to damages or reimbursement of expenses, we shall only be liable insofar as we, our executive personnel or auxiliary staff can be charged with intent or gross negligence. Liability regardless of fault shall remain unaffected in accordance with the Product Liability Act. Liability for culpable violation of fundamental contractual obligations shall also be unaffected; however, apart from cases in the first and second sentence, liability shall be limited to the predictable damages typical for such kind of contract. A change in the burden of proof to the disadvantage of the buyer is not associated with the aforementioned regulations. Liability for guarantees of condition or durability or by maliciously concealing a defect is not affected by the aforementioned provisions

9. APPENDIX

9.1 EMC guidance and manufacturer's declaration

Guidance and manufacturer's declaration – electromagnetic emissions		
The Mobil-O-Graph NG® blood pressure monitor is intended for use in the environment specified below. The purchaser or user of the Mobil-O-Graph NG® blood pressure monitor should ensure that it is used in such an environment.		
Electromagnetic emissions	Compliance	Electromagnetic environment guideline
RF emissions CISPR 11	Group 1	The Mobil-O-Graph NG® blood pressure monitor uses RF energy purely for its own internal function. RF emissions are therefore very low and are not likely to cause any interference in nearby electronic equipment.
RF emissions CISPR 11	Class B	The Mobil-O-Graph NG® blood pressure monitor is suitable for use in all establishments, including domestic establishments and those directly connected to the public low-voltage supply network that supplies buildings used for domestic purposes.

Guidance and manufacturer's declaration – electromagnetic immunity			
The Mobil-O-Graph NG® blood pressure monitor is intended for use in the environment specified below. The purchaser or user of the Mobil-O-Graph NG® blood pressure monitor should ensure that it is used in such an environment.			
Immunity test	IEC 60601-1-test level	Compliance level	Electromagnetic environment – guidance
Electrostatic discharge (ESD) nach IEC 61000-4-2	+/- 6kV contact +/- 8 kV air	+/- 6kV contact +/- 8 kV air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%
Power frequency (50/60Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.

Guidance and manufacturer's declaration – electromagnetic immunity			
The Mobil-O-Graph NG® blood pressure monitor is intended for use in the environment specified below. The purchaser or user of the Mobil-O-Graph NG® blood pressure monitor should ensure that it is used in such an environment.			
Immunity test	IEC 60601-1-test level	Compliance level	Electromagnetic environment – guidance
Radiated RF IEC 61000-4-3	3 V/m 80 MHz to 2,5 GHz	3 V/m	Portable and mobile RF communications equipment should be used no closer to any part of the Mobil-O-Graph NG® blood pressure monitor, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distance: $d = 1.17 \cdot (\sqrt{P})$ P, d = 0.35 $(\sqrt{P})$ P 80 MHz to 800 MHz $d = 0.7 \cdot (\sqrt{P})$ P 800 MHz to 2,5 GHz where p is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in metres (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey, should be less than the compliance level in each frequency range. Interference may occur in the vicinity of equipment marked with the "non-ionising radiation" symbol.

NOTE 1: At 80 MHz and 800 MHz, the higher frequency range applies.

NOTE 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people. Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the Mobil-O-Graph NG® blood pressure monitor is used exceeds the applicable RF compliance level above, the Mobil-O-Graph NG® blood pressure monitor should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as reorienting or relocating the Mobil-O-Graph NG® blood pressure monitor.

Recommended separation distances between portable and mobile RF communications equipment and the Mobil-O-Graph NG® blood pressure monitor			
The Mobil-O-Graph NG® blood pressure monitor is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the Mobil-O-Graph NG® blood pressure monitor can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the Mobil-O-Graph NG® blood pressure monitor as recommended below, according to the maximum output power of the communications equipment.			
Separation distance according to frequency of transmitter in m			
Rate maximum output of transmitter P in W	80 MHz to 800 MHz $d = 1.17 \cdot (\sqrt{P})$	800 MHz to 2.5 GHz $d = 2.33 \cdot (\sqrt{P})$	150 KHz to 80 MHz
0,01	0,12	0,23	not applicabler
0,1	0,37	0,74	not applicable
1	1,17	2,33	not applicable
10	3,70	7,37	not applicable
100	11,70	23,30	not applicable
For transmitters rated at a maximum output power not listed above, the recommended separation distance d in metres (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer. NOTE 1: At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies. NOTE 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.			

Master copy**9.2 Important instructions for the patient**

- The beep signals a measurement. Keep still during the measurement until the process is complete. Let your arm hang loosely or, when sitting, let your lower arm rest on the table or on a support. Avoid moving your hand during the measurement.
- The Start key  triggers an additional measurement. Press this key only if instructed to do so by your doctor (e.g. after taking medication, if you have chest pains or shortness of breath, etc.).
- The Event key  saves the time of an event and triggers a measurement. Press this key only if something important or unusual is happening.
- The Day/Night key  should be pressed immediately before going to sleep and upon waking.
- The ON/OFF key  permits the device to be switched on and off if a measurement is not desired at the set time. To switch off, press the key for longer than 2 seconds.
- A measurement can be broken off at any time by pressing any of the keys. This deflates the cuff.

Warnings:

- The cuff may only be worn on the upper arm and in every case you should ensure that neither the shoulder strap nor the air tube can ever become wrapped around your neck. The air tube should therefore always be worn under the outer clothing (even at night).
- Take care to ensure that the tube of the cuff cannot be kinked or compressed, particularly while sleeping.
- If the measurement causes you pain, switch off the device, take off the cuff and inform your doctor.
- The blood pressure monitor must not come into contact with liquids. Do not wear the equipment in the shower! If, for example, water has entered the device, it may no longer be used. Switch off the device and take out the batteries.
- Serious malfunctions are indicated by a continuous beep. If a continuous beep sound, switch off the device, take off the cuff and inform your doctor.
- The cuff connection (air connector) must always engage with an audible click. Otherwise there will be a leaky connection between the Mobil-O-Graph NG[®] and the tube that will produce incorrect measurements.

9.3 Accessories order form Mobil-O-Graph NG®**Master copy**

ABP monitor Mobil-O-Graph NG®- The long-term blood pressure monitor

Practice:	_____
Street:	_____
Postcode/town:	_____
Tel.:	_____
Fax:	_____

Dealer's stamp:

Please give full address!

- ☐ Please provide a quotation for the following items
- ☐ I wish to order the following items

Item	Number
Cuff S (20-24 cm)	
Cuff M (24-32 cm)	
Cuff L (32-38 cm)	
Cuffe XL (38-55 cm)	
Pouch with Velcro fastening	
Complete cuff tube, 50 cm	
Air connector for cuff tube	
Battery charger	
I.E.M. rechargeable battery AA 1.2 V	
Batteries AA 1.5 V Pack of 4	
PC combi cable serial	
PC combi cable USB	
Combi Handbook	
Infrared component IR-med serial	
Infrared component IR-med USB	
Bluetooth USB dongle	
HM software for Windows CD update	
Event diaries, 3 blocks of 40 sheets	

- ☐ I am interested in a further ABP monitor Mobil-O-Graph NG®
- ☐ in part-exchange for an old unit
- ☐ without part-exchange for an old unit

Date: _____

Signature: _____

9.4 Blood pressure self-monitoring device order form

Master copy



Sabil-O-Graph® / HeartBeat / Klock®

Practice:	_____
Street:	_____
Postcode/town:	_____
Tel.:	_____
Fax:	_____

Dealer's stamp:

Please give full address!

- ☐ Please provide a quotation for the following items
☐ I wish to order the following items

Item	Number
Stabil-O-Graph® - The upper-arm monitoring device (incl. M Cuff)	
Cuff M (24-32 cm)	
Cuff L (32-42 cm)	
HeartBeat - The blood pressure wristwatch:	
in blue	
in yellow	
Klock® - The blood pressure watch	
Infrared interface IR-Med:	
Infrared component IR-med serial	
Infrared component IR-med USB	

Date: _____

Signature: _____